

Asadullah Farooqi (Asad Ullah)

+923111686184, asadullah.itcgcs@gmail.com, [GitHub](#), [LinkedIn](#), [Portfolio](#), Located: **Islamabad, Pakistan**

Languages: **English** (Fluent), **Urdu** (Fluent), **Pashto** (Fluent), **Persian** (Beginner)

Summary: Software engineer with 6+ years of experience developing scalable and robust applications. I've worked in embedded systems, IoT, Web Development, automation, and Fintech. Enthusiastic about Fleet-reliability, Linux, performance, automation, distributed systems, Rust, and Elixir.

SKILLS

Languages: Golang, Python, Bash | **Infrastructure:** Kubernetes, Docker, AWS (EKS, RDS, SQS, SNS, Aurora, etc), Azure IoT, Observability | **Backend:** MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, ETL Pipelines | **Protocols:** gRPC, REST, GraphQL, MQTT, HTTP | **Tools:** Go-kit, Gin, Echo, Django, Flask, Tokio, Git, GitHub Actions

Permanent and Full-Time

Sr Software Engineer at [SafePay](#) 04/2022 - Present

- Added observability in a customized way by adding metrics and better crash-recovery and rollout/rollback components.
- Assist/mentor junior engineers with architecture decisions, making progress in pressured situations
- I implemented a distributed locking mechanism, which prevented concurrent payment capture requests, **resulting in 0 production bugs over 12 months.**
- **Automated 99% of manual reconciliations** by implementing a controllable async Golang service.
- Improved analytics and controlled the RDS cost growth through Elasticsearch integration across the microservices ecosystem with event listeners and backfillers
- To achieve faster backfill for 5+ million records without choking the DB or increasing cost, I wrote jobs using a worker pool concurrency model to sync PostgreSQL and Elasticsearch from the primary database
- Designed and implemented a complete data warehouse infrastructure (PostgreSQL), including ETL pipelines, Kubernetes batch jobs, and SQS event streaming
- Developed an advanced settlement generation system supporting marketplace models (Stripe Connect architecture) for complex multi-party payouts
- Wrote utility billing service covering 90% of national service providers with high availability requirements
- Created new payment gateway integrations (Easypaisa, AbhiPay, card payments) as Golang microservices using Go-kit, gRPC, with optimized MySQL schemas
- Implemented an audit logging system for admin operations, ensuring compliance and full traceability
- Extended core services with security layers: BIN-based MDR charging, validation middleware, gRPC clients, and Protocol Buffer definitions
- Developed payment reporting APIs and complex SQL queries (MySQL/PostgreSQL), powering real-time merchant and admin dashboards
- Tech-stack: Golang, Kubernetes, Docker, AWS (SQS, SNS, RDS, EKS), MySQL, PostgreSQL, Elasticsearch, gRPC, HTTP, ETL

Software Engineer at [Siemens DISW](#) 02/2020 - 03/2022

- Integrated AWS IoT Core and Azure IoT Hub into an edge solution for MQTT telemetry, device provisioning, authentication, and OTA firmware updates. Also added support for Flogo (no-code)
- Developed features for an embedded IoT framework in C for resource-constrained devices
- Built a comprehensive Python test automation framework from scratch for unit, integration, and firmware update testing, reducing manual QA cycles
- Tech: C, Golang, Python, AWS IoT, Azure IoT, Docker, Linux, Make, CMake, MQTT, GCC/Delve Debuggers

Software Engineer Intern at [Maher Inc.](#) 12/2018 - 03/2019

- Fixed UI problems and bugs in [StyleMepretty](#).
- Added APIs for statistics in the backend and charts in the front end to display various matrices. My improvements made the dashboard more analytical, providing clean data and different types of charts.
- Tech-stack: Python, Django, AngularJS, ReactJS, HTML, CSS, Echart JS, Pandas, and Docker.

First Project [Inventory Web App](#) 09/2017 - 03/2018

- Built an inventory-responsive web app for a local businessman to manage branches and ledgers. As my first project, I learned a great deal.
- Tech-stack: Python, Django, [Jinja](#), Bootstrap, [Pythonanywhere](#), Unicorn, WSGI.

EDUCATION

Government College Loralai	FSc (Mathematics)	2013 - 2015
Farooqi Public High School	Matriculation (Science)	2011 - 2013